THE INFORMATION ACCESSIBILITY CRISIS

SIMPEL
SATU LANGKAH SEMUA BANTUAN

8M+

Eligible Indonesian families fail to receive government aid annually.

Sources: World Bank (2023) – only 60% of intended beneficiaries reached; TNP2K (2022) – coverage gap analysis.

- ✓ **Fragmented Rules** – criteria scattered across multiple websites.
- ✓ **High Literacy Demands** – dense legal language, impossible under stress.
- ✓ **Infrastructure Limits** – remote areas, slow 2G, limited smartphones.

Result: Families miss out on support they legally deserve, not because they don't want help, but because the system is too confusing.

SIMPEL—AI Benefits Navigator for Indonesian Families

7 questions, 4 national programs, 3 minutes—SIMPEL (Simple Information System for Understanding Benefit Eligibility) helps Indonesian families find the benefits they may qualify for and what to do next—with explanations, documents, and the right office to visit

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NATIONAL PROGRAM

PKH	BPNT	PIP	PBI-JKN
Cash Transfers	Food Assistance	School Aid	Free Healthcare

INCLUSIVE SOCIAL NAVIGATION

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7 Plain-Language Questions

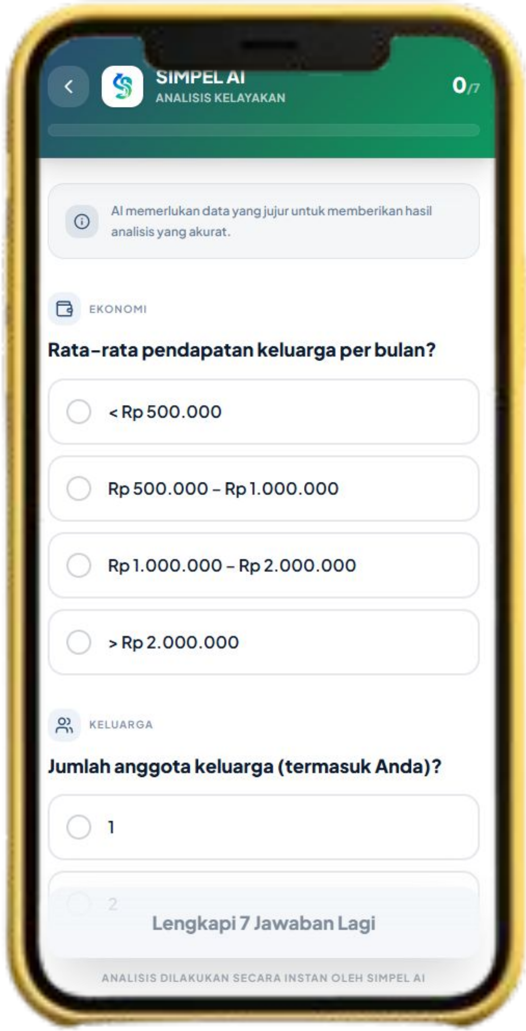
(Bahasa Indonesia & English toggle)

income bracket · family size · school child · toddler/pregnancy · elderly · disability · wall material

No login. No data storage. Zero cost.

Why 7 Questions?

We mapped the eligibility criteria of all four national programs to the minimal set of features that cover every condition — no more, no less. Each question maps directly to one or more program rules.



Designed for Reality

- **Structured multiple-choice** — no typing, reliable on 2G. *Why not a chatbot? Chatbots require typing, stable internet, and tolerance for ambiguity — all of which our user lacks. A fixed questionnaire is deterministic, fast, and works on any connection.*
- **Operated by kader** — community health workers assist families without smartphones.
- **Printable results** — a physical checklist to bring to the village office.

INTERPRETABLE ML vs. RIGID RULES

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🤖 Why Machine Learning?

- Rigid if-else rules break on edge cases (e.g., a family with mixed eligibility signals).
- **A multi-label Random Forest** learns the probabilistic patterns behind real caseworker decisions.

📊 Balanced Training Data

- **4,200+ synthetic households**, including **1,500 negative (ineligible) samples.**
- Fixes discovered bias: wealthy households no longer receive false PKH recommendations.

🖥️ Our Classifier

- **Multi-output Random Forest** (scikit-learn, 100 trees, max-depth=5).
- Single family can be recommended for multiple programs at once.

⏩ Interpretable & Fast

- **Feature importance** explains why a recommendation was made.
- **<10ms inference, <3 MB model, zero external API calls.**
- **Deterministic** — same input always gives same output; we can guarantee it never says "definitely qualify"

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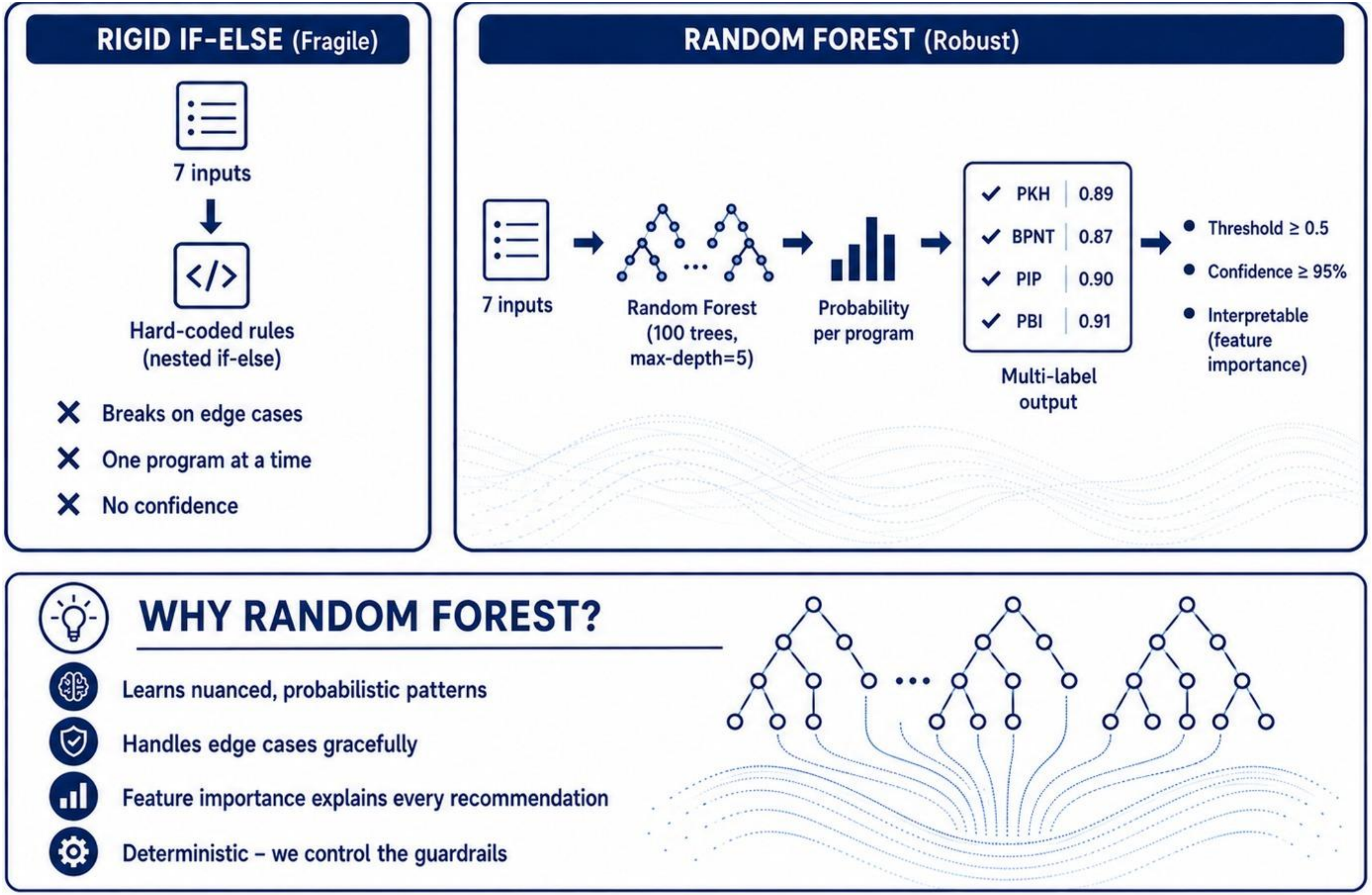

Why Random Forest and not an LLM?

1. **Interpretability** — we can show feature importance to users and judges.
2. **Control** — deterministic output means we can cap confidence and enforce “may qualify” language.
3. **No API costs, no internet needed for the AI inference itself** —the model runs entirely in-memory after the page loads.
4. **Reproducibility** — the entire training pipeline fits in one Jupyter notebook.

RESPONSIBLE AI & FAIRNESS DEBUGGING

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The Challenge We Caught

Our first model gave 80% PKH confidence to a wealthy family with brick walls and no dependants. This was invisible in aggregate metrics; we found it by manually testing extreme cases.

Our Fix — Why It's Rigorous

- **+1,500 negative training samples** — families who clearly don't qualify
- **Reduced label noise** on dominant classes (5 % → 0.5 %)
- **Confidence capped at 95 %** — never "100 % certain"
- All outputs use "you may qualify," never certain

Privacy by Design

Zero-retention ephemeral state. No data stored, logged, or transmitted. Page refresh = complete wipe.

- **Why ephemeral?** Our users often share devices or use them in public spaces. Storing any data would be a liability. We chose to store nothing at all — privacy isn't a feature, it's the default.

- **Why a static model if rules change?** We are transparent about this limitation. The model is trained on current rules. Roadmap: a DTKS sync + guardrailed chatbot layer for up-to-date context — while the core model remains safe, deterministic, and offline-capable.

USER-CENTERED DESIGN IN ACTION

Step 1 – Ephemeral Questionnaire

Dropdowns/radio buttons in Bahasa Indonesia (English toggle). No typing, no personal data. This is what the kader sees when she opens SIMPEL to help Ibu Sari — seven questions, under three minutes.

Step 2 – Explicit Disclaimers

Persistent yellow banner: "This is not an official decision. Please verify with the local village office.." The AI never decides — a human caseworker at the village office does.

Step 3 – Actionable Output

Expandable cards with:

- Program name + confidence ≤95 %
- "Mengapa?" — plain-language reason
- Document checklist (KTP, KK, SK)
- Exact office to visit (Dinas Sosial, kelurahan, school, BPJS branch)
- Link to official source (Kemensos, Kemendikbud, BPJS)
- Print-friendly (Ctrl + P)

Why human-in-the-loop? The AI suggests; a human verifies. No machine should make life-altering enrollment decisions. We deliberately excluded any pass/fail verdict.

SCALE, IMPACT, and FUTURE ROADMAP

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Why Each Roadmap Item, and How We'll Build It

VOICE INPUT
Web Speech API (online) or Vosk (offline) converts speech to text; the existing feature pipeline remains unchanged.

Multi-language
The dictionary-based language toggle (already built) extends to Javanese, Sundanese, etc.

WhatsApp/SMS
A lightweight Flask endpoint triggered after prediction sends results via WhatsApp Cloud API; no browser needed.

100% offline
Compile the exact Random Forest to JavaScript via m2cgen, bundle with a PWA manifest — inference runs on-device, zero server.

DTKS sync
A periodic local SQLite cache of Indonesia's official welfare database enables offline eligibility verification without live API calls.

Guardrailed Q&A chatbot
A small LLM (e.g., Gemma 2B) sandboxed with a system prompt: "Only answer questions about programs, documents, and procedures." It explains — it never decides.

IMPACT

If adopted by just 1 % of Indonesia's 70,000 kader, SIMPEL could screen over 2 million families annually — reducing per-family admin time from hours to under three minutes.

THANK YOU

Reference

- *World Bank (2023) – only 60% of intended beneficiaries reached; TNP2K (2022) – coverage gap analysis.*
- *https://thumbs.dreamstime.com/b/asian-village-old-people-learning-laptop-technology-rural-area-india-january-asian-village-old-people-learning-169188246.jpg*
- *https://asianews.network/wp-content/uploads/2025/06/AFP__20250507__449K9T8__v1__Preview__IndonesiaPovertyEconomy.jpg*